

Mini Test 3 Pathway 1 and 2
Quadratics



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Resources allowed: Calculator and 1 x A4 page one side

Name: Solutions

Class: _____

Time: 45 minutes

Total Marks: 46

Part A:

1. [10 marks: 1, 1, 2, 2, 2, 2]

Expand the following, simplify if possible:

a) $2(x+3)$ $\left(\frac{1}{2}\right) \left(\frac{1}{2}\right)$
 $= 2x + 6$ ✓

b) $2x(y+7)$ $\left(\frac{1}{2}\right) \left(\frac{1}{2}\right)$
 $= 2xy + 14x$ ✓

c) $(h+3)(h+4)$
 $= h^2 + 4h + 3h + 12$ ✓
 $= h^2 + 7h + 12$ ✓

d) $(d+8)(d-8)$
 $= d^2 - 8d + 8d - 64$ ✓
 $= d^2 - 64$ ✓

e) $(f+3)^2$
 $= f^2 + 3f + 3f + 9$ ✓
 $= f^2 + 6f + 9$ ✓

f) $(2j-3)^2$
 $= 4j^2 - 6j - 6j + 9$ ✓
 $= 4j^2 - 12j + 9$ ✓

2. [8 marks: 1, 1, 2, 2, 2]
 Fully Factorise the following:

a) $2q^4$ $\left(\frac{1}{2}\right) \left(\frac{1}{2}\right)$
 $2(q-2)$ ✓

b) $6r+3r^2$ $\left(\frac{1}{2}\right) \left(\frac{1}{2}\right)$
 $3r(2+r)$ ✓

c) $c^2+8c+15$
 $(c+3)(c+5)$ ✓✓

d) a^2-9
 $(a+3)(a-3)$ ✓✓

e) $x^2-12x+36$
 $(x-6)(x-6)$ ✓✓

3. [7 marks: 1, 2, 2]

Solve the following quadratics:

a) $(d+3)(d+3)=0$

$$d = -3 \checkmark$$

b) $(w+3)(w+4)=0$

$$x = -3 \checkmark \text{ or } x = -4 \checkmark$$

c) $y^2 - 3y - 10 = 0$

$$(y-5)(y+2) \checkmark$$

$$y = 5 \text{ or } y = -2 \checkmark$$

$g^2 - 4g = 12$

$$g^2 - 4g - 12 = (g-6)(g+2) \checkmark$$

$$g = 6 \text{ or } g = -2 \checkmark$$

Part B: A and B Level Questions

4. [6 marks: 1, 1, 1, 1, 2]

Factorise the following:

a) $t(t-4) + 3(t-4)$

$$(t+3)(t-4) \checkmark$$

b) $h^2 + 2hk + k^2$

$$(h+k)^2 \checkmark$$

c) $2x^2 + 9x + 10$

$$(x+2)(2x+5) \checkmark$$

d) $4t^2 + 12t + 8$

$$4(t+1)(t+2) \checkmark$$

e) $25 - (m+n)^2$

$$5^2 - (m+n)^2 = [5 + (m+n)][5 - (m+n)] \checkmark$$

$$= (5+m+n)(5-m-n) \checkmark$$

5. [2 marks]

Solve the following quadratic: $d^2 - 25 = 0$

$$(d+5)(d-5)$$

$$d = 5 \text{ or } d = -5 \checkmark \checkmark$$

6. [6 marks: 2, 2, 2]

Solve the following quadratics by completing the square:

a) $d^2 + 6d + 2 = 0$

$$\begin{aligned} d^2 + 6d &= -2 \\ d^2 + 6d + 9 &= -2 + 9 \\ (d+3)^2 &= 7 \\ d+3 &= \pm\sqrt{7} \\ d &= -3 \pm \sqrt{7} \end{aligned}$$

b) $p^2 - 3p - 4 = 0$

$$\begin{aligned} p^2 - 3p &= 4 \\ p^2 - 3p + (1.5)^2 &= 4 + 1.5^2 \\ (p-1.5)^2 &= 6.25 \\ p-1.5 &= \pm\sqrt{6.25} \\ p &= 1.5 \pm 2.5 \\ p &= 4 \text{ or } p = -1 \end{aligned}$$

c) $g^2 - 10g + 1 = 0$

$$\begin{aligned} g^2 - 10g &= -1 \\ g^2 - 10g + 25 &= -1 + 25 \\ (g-5)^2 &= 24 \\ g-5 &= \pm\sqrt{24} \\ g &= 5 \pm \sqrt{24} \end{aligned}$$

7. [7 marks: 1, 1, 2, 2, 1]

This diagram shows two squares whose sides are a metres and b metres.

a) What is the area of the larger square?

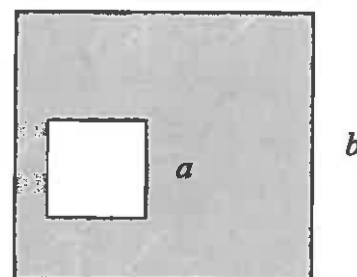
$$b^2$$

b) What is the area of the smaller square?

$$a^2$$

c) Write an expression for the area of the shaded region.

$$b^2 - a^2$$



d) Factorise the expression you wrote in part C.

$$(b+a)(b-a)$$

e) If $a = 37\text{m}$ and $b = 137\text{m}$, use your answer from part d to find the area of the shaded region.

$$(137+37)(137-37) = 17400 \text{ m}^2$$

End Test

